Problem Statement

Atomberg’s current water-purifier portfolio already combines RO, UF, UV and Alkaliser stages, with smart TDS-based modes and IoT connectivity. The AtomQUEST brief explicitly asks participants to push this architecture further, particularly through water-quality sensing beyond TDS and data-driven filter-failure prediction. AQUASENSE-X addresses this gap by creating a closed-loop purification system that measures the incoming water, selects an appropriate treatment path, verifies the output, and continuously estimates the health of its consumable components.

Proposed Idea / Mechanism

AQUASENSE-X adds a multi-parameter sensing layer around the existing purification architecture. Instead of using TDS as the primary proxy for water quality, the prototype combines TDS/conductivity, pH, turbidity, temperature, flow and pressure measurements to characterize water conditions and treatment performance. A controller uses these signals to select or tune UF/RO/UV operation and monitor operating conditions. A second sensing layer verifies the treated-water state before dispensing. Operational data—including litres processed, input-water characteristics, pressure and treatment cycles—is logged to estimate filter degradation and provide a predicted replacement window rather than relying only on calendar-based replacement.

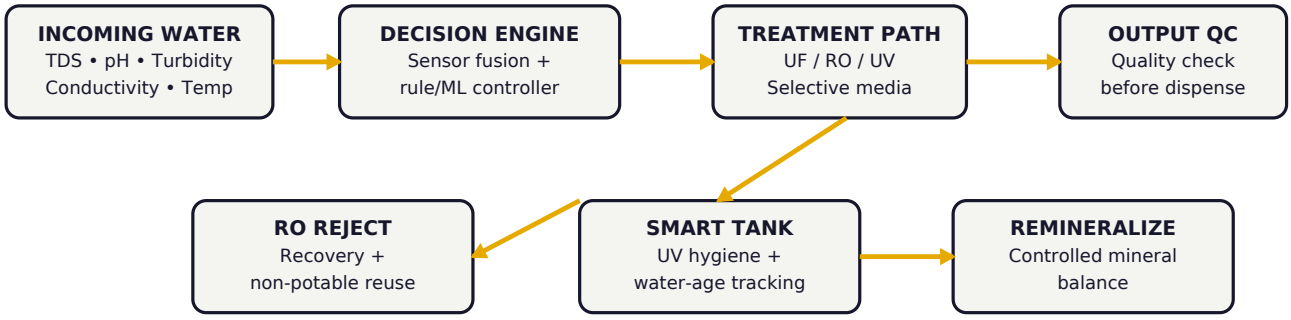
Design Targets (prototype targets; to be validated experimentally)

Metric	Current / Baseline	AQUASENSE-X Target
Water-quality decision	Primarily TDS-based	≥5 measurable parameters used in control logic
Output verification	Treatment-stage monitoring	Independent post-treatment quality check before dispense
Filter maintenance	Time / usage based	Predictive estimate with target ±10% error after calibration datas
RO reject handling	Conventional reject discharge	≥50% of reject stream routed to controlled non-potable reuse in
Safety response	Fault alert / service action	Fail-safe dispense lock when critical sensor conditions are out of

Important: The numerical values above are engineering targets for the prototype, not claims of validated performance.
Final figures will be established through controlled testing against defined water samples.

C · TECHNICAL FEASIBILITY

System Architecture



Closed-loop architecture: sense → decide → purify → verify → learn

The proposed architecture is technically feasible because it builds on established water-treatment subsystems while adding a practical sensing and control layer. The inlet stage measures water condition and hydraulic behaviour. A microcontroller or embedded Linux controller performs sensor fusion and treatment decisions. The treatment path retains proven unit operations—sediment/carbon pre-treatment, UF, RO and UV—with optional selective media where a specific contaminant profile justifies it. A post-treatment sensor layer checks process conditions before water reaches the smart tank or dispensing point.

Core Components & Subsystems

Subsystem	Prototype Components / Function
Sensing	TDS/conductivity sensor, pH sensor, turbidity sensor, temperature sensor, flow meter, pressure sensors
Controller	ESP32 / STM32-class MCU; optional Raspberry Pi-class gateway for data logging and analytics
Treatment	Sediment + activated carbon, UF membrane, RO membrane/pump, UV module; optional selective adsorption stage
Hydraulics	Solenoid valves, check valves, flow restrictor, pressure regulator, tubing and fittings
Verification	Post-treatment TDS/conductivity, turbidity and flow/pressure checks; UV operating-state monitoring
Analytics	Time-series database/logging, feature extraction and a lightweight regression/anomaly model for filter health
User interface	Local display / mobile dashboard showing water status, filter health, alerts and maintenance guidance

Safety-Critical Design

The prototype will be designed so that sensing and analytics do not bypass physical treatment safeguards. A fail-safe state will prevent dispensing when critical process conditions are invalid—for example, abnormal pressure/flow, failed sensor plausibility checks, or a treatment-stage fault. UV effectiveness will be monitored through lamp/driver status and, where feasible, intensity feedback. The system will clearly distinguish measured parameters from laboratory-only contaminant verification; it will not claim that TDS, pH or turbidity sensors alone identify specific contaminants such as lead, arsenic or PFAS.

Technical principle: sense → classify operating condition → select treatment → verify output → log degradation → predict maintenance. This creates a closed-loop purifier rather than simply adding more filtration stages.

D · EXECUTION FEASIBILITY

Rough Bill of Materials / Components

Category	Indicative Items	Order of Magnitude
Sensors	TDS/EC, pH, turbidity, temperature, flow, pressure	■8k–■20k
Treatment	UF + RO membrane, RO pump, UV, sediment/carbon stages	■10k–■20k
Control	MCU, ADC/interface boards, relays/MOSFETs, display	■3k–■8k
Hydraulics	Valves, fittings, tubing, check valves, regulator	■4k–■10k
Analytics	Gateway / storage / prototype dashboard	■2k–■8k
Mechanical	Prototype enclosure, tank integration, mounts	■5k–■12k
Estimated prototype total	Depends on available Atomberg hardware and custom fabrication	≈ ■35k–■80k

Component Sourcing Risk

Most sensing, control and hydraulic components are commercially available in India. The principal risks are calibration-quality sensors, food-contact-compatible fittings/tubing, UV modules with dependable specifications, and any custom mechanical interface required to integrate the prototype with an Atomberg platform. No single custom semiconductor is required. If a specialized contaminant-selective cartridge is explored, it will be treated as an optional module rather than a dependency for the first prototype.

Build Plan to Round 2

Phase	Work	Deliverable
1 · Bench setup	Assemble sensor board and characterize sensor repeatability using controlled water samples.	Calibrated sensor stack
2 · Hydraulic prototype	Integrate valves, flow meter, pressure sensing and existing filtration stages.	Working water-flow loop
3 · Closed-loop control	Implement treatment-state logic, fault handling and output verification.	Level 1 functional prototype
4 · Data collection	Run repeated cycles across varied TDS/turbidity/pH and operating conditions; log filtering usage.	Training/validation dataset
5 · Prognostics	Build lightweight filter-health model and maintenance prediction dashboard.	Predictive maintenance demo
6 · Validation	Measure recovery, output quality indicators, fault response and prediction error.	Quantified prototype results

Expected Impact

AQUASENSE-X aims to make purification more adaptive, transparent and maintainable. By sensing beyond TDS, verifying treatment performance and predicting filter degradation from operating data, the system can reduce unnecessary treatment, reduce avoidable maintenance, improve user visibility into purifier health, and encourage responsible handling of RO reject water. The concept directly aligns with the AtomQUEST 2026 Water Purifier challenge and is intended to culminate in a live, working prototype rather than a slide-only concept.

Source alignment: The AtomQUEST 2026 brief states that the Water Purifier track seeks solutions around sensing beyond TDS, water enhancement, DIY filter replacement and data-driven filter-failure prognostics. This proposal prioritizes the first and fourth areas while extending them into a unified closed-loop architecture.